AU Property Insights

Power BI Dashboard — Insights Summary

Task 8 — BI Advanced

File: AUProperty_PowerBI_Dashboard.pbix

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1. Headline Metrics

Figures below are computed directly from the AUPropertyDW model (all suburbs, unfiltered) and match the Summary page KPI cards.

Metric	Value
House listings tracked	4,996
Average house value	$563,216
Median house value	$440,000
Rental listings tracked	14,719
Average weekly rental	$393
Distinct schools	6,443 (6,573 presence records)
Distinct transport stops	9,689 (12,060 stop records)
Total recorded crime incidents	10,900,021
Distinct offence categories/subcategories	112

How to read these numbers: Figures throughout this document are portfolio-wide (all suburbs/states combined). Individual report pages let you slice by State/City, so a filtered view — e.g. Victoria only shows $700,716 average / $567,500 median house value on 719 listings — will legitimately differ from the numbers below. Compare like-for-like filter context before treating two numbers as inconsistent.

2. House Value Insights

The average house value ($563K) sits well above the median ($440K) — a classic right-skewed distribution, confirmed by the price-band breakdown: listings cluster heavily in the lower-to-mid bands and taper off sharply at the top end.

Price band	Listings	Share
Under $300K	841	17%
$300K – $450K	1,706	34%
$450K – $600K	854	17%
$600K – $800K	800	16%
$800K – $1.2M	444	9%
$1.2M – $2M	294	6%
Above $2M	57	1%

Nearly two-thirds of all listings (68%) fall under $600K, while only 7% exceed $1.2M — the top end of the market is a small but visible long tail rather than the norm.

2.1 Highest-value cities

- Armadale reports the single highest average house value ($2.0M), but this is based on just one listing in the warehouse — an outlier, not a market trend.
- Among cities with meaningful listing volume, Mornington ($869.5K), Melbourne ($730.9K) and Wollongong ($698.9K) are the genuine top performers.

3. Rental Value Insights

Weekly rent band	Listings	Share
Under $250/wk	2,063	14%
$250 – $350/wk	4,983	34%
$350 – $450/wk	3,934	27%
$450 – $600/wk	2,343	16%
$600 – $900/wk	1,054	7%
$900 – $1,500/wk	305	2%
Above $1,500/wk	37	<1%

The rental market shows the same right-skewed shape as house values: 61% of listings sit between $250–$450/wk, and only 2% exceed $900/wk.

- Armadale again tops the average-rent ranking ($691.5/wk), but on only 6 listings — a small sample, not a stable signal.
- Among higher-volume cities, Nowra ($550/wk), Wollongong ($459/wk) and Tweed Heads ($452/wk) lead.

4. Schools & Transport Insights

4.1 Schools

6,443 distinct schools are represented across 6,573 presence records, spread across every state in the warehouse, with Melbourne, Adelaide and Sydney showing the highest school counts by city.

4.2 Transport

Scope note: Every one of the 12,060 transport stop records in the warehouse is tagged Mode = “train”. There is no bus, tram or ferry data in the current source extract, so the Transport page reports rail-stop coverage only, not all-mode public transport access.

Transport-stop coverage is uneven across states:

State	Stop records	Distinct stops
Victoria	6,297	5,495
South Australia	5,079	3,619
New South Wales	613	508

New South Wales — home to Sydney, Australia's largest city — shows far fewer transport-stop records than Victoria or South Australia. This is almost certainly a data-coverage gap in the source transport file rather than a genuine finding about NSW public transport access, and should be read as a limitation rather than an insight.

5. Crime Insights

State	Total recorded incidents
New South Wales	8,697,040
Victoria	1,651,763
South Australia	551,218

New South Wales accounts for roughly 80% of all recorded incidents, and Sydney alone (5.64M) reports more than four times the next-highest city, Melbourne (1.28M).

Interpretation caveat: The dashboard reports total recorded incidents, not a rate per capita or per suburb. NSW's very high total most likely reflects the completeness/granularity of the source crime dataset for that state, combined with Sydney's population size — not necessarily a higher crime rate. Cross-state or cross-city comparisons should be read with this caveat rather than as a like-for-like rate comparison.

5.1 Top offence categories

Offence category	Incidents
Theft	2,728,581
Transport Regulatory Offences	1,094,119
Property and Deception Offences	978,506
Against Justice Procedures	889,491
Malicious Damage to Property	874,026
Assault	835,987

Theft is the single largest offence category, at roughly 25% of all recorded incidents — consistent with theft and regulatory/procedural offences dominating overall volumes across every state.

6. Visualization Choices

Every detail page (House Value, Rental Value, Schools, Transport, Crime) follows the same layout logic, chosen to match what each visual type communicates best:

- KPI cards (top row) — single headline numbers (average/median/count) need no chart; a card gives the answer in under a second and anchors the rest of the page.
- Map (filled/bubble map on Suburb, sized or coloured by the metric) — house value, rental value, school, transport and crime data are all inherently geographic, so a map is the fastest way to spot regional concentration that a table or bar chart would hide.
- Horizontal bar chart, "<Metric> by City" — city names are long text labels; horizontal bars keep them readable and make ranking cities against each other a simple length comparison, which a column chart or line chart would not do as clearly for a categorical, non-time dimension.
- Bucket/band column chart (House Value, Rental Value pages) — turns a continuous numeric field into a distribution shape (see Section 7), answering "where does the market cluster?" rather than just "what's the average?".
- Donut chart (Schools page, School Count by SchoolType) — SchoolType is a small, low-cardinality category; a donut shows each type's share of the whole at a glance, which is the donut chart's strength over a bar chart when the question is about proportion rather than precise ranking.
- Detail table (Schools, Crime pages) — kept alongside the charts for users who need the exact underlying rows/values a chart can't show precisely (e.g. exact incident counts by OffenceSubcategory).
- Summary page — one KPI card + one "by State" bar chart per analysis area, so all five areas can be compared side-by-side on a single screen; drill-through (Section on Data Connection Documentation) then lets a user jump from that high-level view into the matching full detail page.
- Slicers (State, City, and a page-specific dimension such as OffenceCategory or RentalHouseType) — placed on every detail page so a user can narrow from national to city-level without leaving the page, rather than needing a separate page per state.

7. Bucketing and Grouping Approach

House values and rental amounts are continuous numeric fields, so a plain bar/column chart on the raw value would produce thousands of single-value bars and no usable pattern. Both were converted into fixed price/rent bands using the same technique:

- A calculated column (HouseValue Bucket / RentalValue Bucket) uses SWITCH(TRUE(), ...) to test the raw value against a set of ascending thresholds and return a text label for the matching band (e.g. "$300K - $450K").
- A second, hidden calculated column (HouseValue Bucket Sort / RentalValue Bucket Sort) returns the same band as an integer 1–7 instead of text, and is set as the text column's Sort by Column. Without this, Power BI would sort the bands alphabetically ($1.2M-$2M before $300K-$450K) or by count instead of by ascending price — the hidden sort column forces the correct logical order.
- The band widths themselves were chosen to give a roughly even number of bands across the observed value range while keeping thresholds at intuitive round numbers ($300K, $450K, $600K, $800K, $1.2M, $2M for house value; $250, $350, $450, $600, $900, $1,500/wk for rent), rather than using equal-width or equal-count (quantile) bins.

The exact DAX for both calculated columns is included in the Data Connection Documentation (Section 8.1).

8. Data Quality Notes

- Three suburbs — UNKNOWN, WALL FLAT and RETREAT — carried invalid or placeholder coordinates that plotted them onto Europe/Africa on the geographic visuals. They are excluded from every page via a report-level filter (“Filters on all pages”) so they don't distort totals or maps.
- A small number of DimSuburb rows had a text-encoding artefact (“?Alice Springs” instead of “Alice Springs”), corrected in the Power Query load step — see the Data Connection Documentation for details.
- Some records carry City = “Unknown” or State = “Unknown”; these remain in the model (they are not part of the three excluded suburbs) and should be treated as missing/unclassified rather than a distinct real location when reading city- or state-level breakdowns.